

Abdalla Abdelmagid

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EDUCATION

Towson University, Towson, MD

B.S. Computer Science — May 2025

GPA: 3.8/4.0

- **Key Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Mobile App Development, Software Quality Assurance & Testing, Object-Oriented Programming, Web-Based Programming
- **Honors:** Cum Laude, 8x Dean's List

TECHNICAL SKILLS

Languages: Python, Swift, Java, JavaScript/TypeScript, C++, SQL, Go, HTML/CSS

Frameworks & Tools: SwiftUI, React, Node.js, Express, Spring Boot, JUnit, Docker, AWS, Firebase, Git/GitHub, CI/CD

Data & Analytics: pandas, NumPy, Tableau, PyTorch, TensorFlow, Matplotlib

EXPERIENCE

Independent iOS Developer

Remote

Self-Employed

Jan 2025 – Present

- Shipped 14 native iOS apps to the App Store, reaching over 10,000 downloads and maintaining a 4.6 average rating.
- Took apps through the entire lifecycle, from concept and Figma design to development, testing, and successful App Store launches.
- Monetized apps through subscriptions and in-app purchases, generating over \$1000 in monthly recurring revenue.

Data Science Research Fellow

Baltimore, MD

National Science Foundation (NSF)

Sep 2024 – Dec 2024

- Analyzed Baltimore City crime/resource datasets with Python (pandas) and SQL, applying regression and clustering to uncover correlations.
- Built interactive Tableau dashboards and Matplotlib visualizations to highlight key trends.
- Presented insights and policy recommendations to NSF and university stakeholders.

Software Engineer Intern

Remote

Revature

May 2024 – Aug 2024

- Engineered 6 RESTful API endpoints in Java/Spring Boot with JWT authentication, securing application data for 1,000+ users.
- Optimized database queries with Spring Data JPA, reducing latency by 30% and improving platform responsiveness.
- Authored 15+ unit/integration tests with JUnit 5 and Mockito, uncovering recurring bugs and preventing crashes.

Software Engineer Intern

Columbia, MD

HQLabs

May 2023 – Aug 2023

- Built a real-time transaction monitoring dashboard with React, TypeScript, and Tremor, enabling live financial data visualization.
- Developed high-throughput gRPC services in Go to process financial transactions, improving reliability and throughput.
- Deployed microservices with Terraform, streamlining CI/CD and reducing deployment errors.

PROJECTS

University Marketplace

Feb 2025

- Built secure campus marketplace (MERN, Firebase Auth) enabling verified .edu sign-ups, real-time chat, and Stripe-powered payments.
- Designed flexible MongoDB schemas and optimized queries to ensure smooth loading and filtering of marketplace items.

nVolve — Event Discovery App

Nov 2024

- Developed SwiftUI app displaying geolocated events; used Alamofire + Codable for REST API consumption and Core-Location for GPS services.
- Implemented a MapKit-based interface with filtering and favorites features to help users find and bookmark nearby events.

Movie Recommender Web App

May 2024

- Collaborated in an Agile team of 4 to design a responsive UI (HTML/CSS/JS) and build a RESTful backend (Java, Spring) interfacing with MySQL and OpenAI GPT-3 to suggest films.
- Implemented JWT-based authentication and deployed the application in a Dockerized environment for public access.